

Anish Badri R S

CS ENTHUSIAST

Chennai, Tamil Nadu, India

☎ (+91) 96001 98790 | ✉ anishbadhri@gmail.com | 🏠 anishbadhri.github.io | 📷 anishbadhri | 🌐 anishbadri

“Do or not do, there is no TODO.”

Experience

Google

Bangalore, India

SOFTWARE ENGINEER

Aug. 2020 - Present

- **Tech Stack:** Java, Typescript, Cloud Spanner, Message Queues, ETL Pipelines, Data Migration
- Tech lead in the Customer Onboarding team, built multiple products to migrate data from third-party services to Google.
- Led the end-to-end design of full-stack data migration and identity synchronization services for enterprise customers.
- Onboarded over **100K users** from various sources including Microsoft, IMAP servers and Dropbox.
- Drove significant performance optimizations in migration services, enhancing scalability and reducing API latency by up to **67%**.
- Experienced in Java and Typescript with expertise in full-stack development, distributed systems, and databases.

Motorq

Chennai, India

SOFTWARE ENGINEERING INTERN

Dec. 2019 - Jan. 2020

- **Tech Stack:** Azure Snowflake, Azure Blob Storage, Node.js, Typescript
- Automated A-B testing with data returned by production code & new code.
- Suggested improvements to testing method based on the stable marriage problem.
- Reduced time taken for pushing code to production by 12 hours.

Google India Pvt. Ltd.

Bangalore, India

SOFTWARE ENGINEERING INTERN

May. 2019 - Jul. 2019

- **Tech Stack:** Java, Cloud Spanner, Message Queues
- Built a bridge from Index Pagination to Token Pagination for the list users API call, making time per call independent of the number of users.
- Reduced upper-bound to 3s per call from the earlier upper-bound of 12s.
- Worked with technologies such as Cloud Spanner & Protocol Buffer.

Freshworks

Chennai, India

SOFTWARE DEVELOPMENT INTERN

May. 2018 - Jul. 2018

- **Tech Stack:** Node.js, SDK Development
- Integrated the flagship product of Freshworks (Freshdesk) with third party applications using Freshworks SDK running on Node.js.
- Improved productivity by preventing tab-switches & allowing data to be synchronised across applications.

Projects & Research Papers

Complyer

Chennai, India

SKILLS USED: C++, CMAKE, ANTLR, Z3, MODEL CHECKING

Dec. 2019 - May. 2020

- Implemented a compiler to verify whether a model satisfies a given property.
- Verifies for both safety & temporal properties specified using propositional logic & linear temporal logic respectively.
- Uses an SMT solver to verify the model. The algorithms used included k-Induction, LTL Bounded Model Checking & IC3.

Complyer-SAT

Chennai, India

SKILLS USED: C++, CMAKE, BOOLEAN SATISFIABILITY

Jul. 2020

- Devised an application to determine if there exists an assignment of variables that satisfies a given Boolean formula.
- Implements Tseytin transformation to convert expression to CNF.
- The solver runs using DPLL algorithm. Optimizations done are 2 - SAT & HORNSAT.

Graph Recommendation Tool

Chennai, India

SKILLS USED: PYTHON

Apr. 2020 - May. 2020

- Devised a command-line application and accepts the template & input graphs and normalizes the input graph to best fit the template graph.
- Normalisation can be done through comparison of only slope or of slope & position of each point.
- Allows an error margin & learning rate to be configured on runtime.

Bounded Model Checking of MFOTL Properties

Chennai, India

SKILLS USED: MODEL CHECKING

Oct. 2018 - Nov. 2018

- Specification and verification of infinite state space systems using a language which has Monadic First-Order (MFO) sentences.
- The MFO sentences return a bound which can be used to restrict the state space of the input system, thereby making the verification problem decidable.

Graph Database Modelling of College Data

Chennai, India

SKILLS USED: NODEJS, NEO4J, GRAPHQL

Jul. 2020

- Designed a scalable graph database schema for easy storage & access of various data based on a variety of use cases.
- Proposed a back-end implementation for an API server to access data from said database.

Command Line Tools for Codeforces

Chennai, India

SKILLS USED: PYTHON, BEAUTIFULSOUP

Nov. 2019

- Built multiple command line tools to perform functions such as problem recommendation, downloading sample input & output, fetch unsolved virtual contest list and recent list of problems solved for the online judge, Codeforces.

Skills

Programming	C++, Java, Typescript, SQL, Python, Shell Script, NuSMV
Concepts	Data Structures, Algorithms, Boolean SATisfiability, Model Checking
Databases	Neo4j (Graph Database), MySQL (Relational Database), Cloud Spanner
Frameworks	Git, Protocol Buffers, Docker, Node.js, CMake, REST, ANTLR, GraphQL
Languages	English, Tamil

Education

SSN College of Engineering (affiliated to Anna University)

Chennai, India

B.E. IN COMPUTER SCIENCE AND ENGINEERING

Jul. 2016 - Jun. 2020

CGPA: 7.8/10.0

Certifications

2020	Neo4j Certified Professional , Certified in Graph Databases	Neo4j
2020	Data Structures & Algorithms , 6 courses; Average Grade: 98.54%	Coursera
2020	Introduction to Discrete Mathematics for Computer Science , 5 courses; Average Grade: 100%	Coursera
2017	Design and Analysis of Algorithms , Grade: 81%; Top 5% in the nation	NPTEL
2017	Model Checking , Grade: 84%	NPTEL
2017	Programming, Data Structures & Algorithms using Python , Grade: 81%	NPTEL

Achievements

PROFESSIONAL

2020-	Leadership Bonuses , Awarded with >10 bonuses for exemplary work.	Google
2020-	Peer Bonuses , Awarded with >30 bonuses for supporting team members.	Google

ACM ICPC & PROVINCIALS

2019	Amritapuri Doublesite Regionals , Ranked 61 out of 324 teams as a member of team IC3.	ICPC
2019	Gwalior-Pune Regionals , Ranked 64 out of 113 teams as a member of team IC3.	ICPC
2019	Chennai Provincials , Ranked 3rd as a member of team IC3.	Provincials
2017	Kharagpur Regionals , Ranked 51 out of 64 teams as a member of team Code_Overload.	ICPC
2017	Chennai Provincials , Ranked 4th as a member of team Code_Overload.	Provincials

ACADEMIC

2018	eXLog Math Quiz , Won 1st place out of 200 teams in the inter-state math quiz.	SSN
2018	Smart India Hackathon , Finalist as a member of team Tandem Felix.	SIH
2018, 19	Facebook Hackercup , Qualified for Round 2	Facebook
2018	Google Codejam , Qualified for Round 2	Google
2017	Codechef Snackdown , Qualified for Elimination Round	Codechef
2016-20	Technical Symposiums , Won in multiple technical events held at various inter-college symposiums.	
2016	Indian National Olympiad in Informatics , Qualified by clearing Zonal Informatics Olympiad.	IARCS

Organizations & Interests

Competitive Programming	Active Problem Solving on Codeforces and Codechef.
SSN Coding Community	Head of SSN Coding Community - MAXFLOW.